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 Studierichting: Informatica
 Oefeningenreeks 7B nr. 2.1

$$w = \ln(x^2 + y^2 + z^2) \quad x = s - t \quad y = s + t \quad z = 2\sqrt{st}$$

$$\frac{dx}{ds} = 1 \quad \frac{dx}{dt} = -1$$

$$\frac{dy}{ds} = 1 \quad \frac{dy}{dt} = 1$$

$$\frac{dz}{ds} = \frac{2t}{2\sqrt{st}} \quad \frac{dz}{dt} = \frac{2s}{2\sqrt{st}}$$

$$\frac{\partial w}{\partial x} = \frac{2x}{x^2 + y^2 + z^2}$$

$$\frac{\partial w}{\partial y} = \frac{2y}{x^2 + y^2 + z^2}$$

$$\frac{\partial w}{\partial z} = \frac{2z}{x^2 + y^2 + z^2}$$

$$\begin{aligned} \frac{\partial w}{\partial s} &= \frac{2x\sqrt{st}}{(x^2 + y^2 + z^2)\sqrt{st}} + \frac{2y\sqrt{st}}{(x^2 + y^2 + z^2)\sqrt{st}} + \frac{2zt}{(x^2 + y^2 + z^2)\sqrt{st}} \\ &= \frac{2x\sqrt{st} + zy\sqrt{st} + 2zt}{(x^2 + y^2 + z^2)\sqrt{st}} \\ &= \frac{2(s-t)\sqrt{st} + 2(s+t)\sqrt{st} + 4t\sqrt{st}}{((s-t)^2 + (s+t)^2 + (2\sqrt{st})^2)\sqrt{st}} \\ &= \frac{2s\cancel{2t} + 2s\cancel{2t} + 4t}{s^2\cancel{2st} + t^2 + s^2\cancel{2st} + t^2 + 4st} \\ &= \frac{4t}{2(s+t)^2} \\ &= \frac{2}{s+t} \end{aligned}$$

$$\begin{aligned} \frac{\partial w}{\partial t} &= \frac{-2x\sqrt{st}}{(x^2 + y^2 + z^2)\sqrt{st}} + \frac{2y\sqrt{st}}{(x^2 + y^2 + z^2)\sqrt{st}} + \frac{2zs}{(x^2 + y^2 + z^2)\sqrt{st}} \\ &= \frac{-2x\sqrt{st} + zy\sqrt{st} + 2zs}{(x^2 + y^2 + z^2)\sqrt{st}} \\ &= \frac{-2(s-t)\sqrt{st} + 2(s+t)\sqrt{st} + 4s\sqrt{st}}{((s-t)^2 + (s+t)^2 + (2\sqrt{st})^2)\sqrt{st}} \\ &= \frac{\cancel{2s} + 2t\cancel{2s} + 2t + 4s}{s^2\cancel{2st} + t^2 + s^2\cancel{2st} + t^2 + 4st} \\ &= \frac{4(s+t)}{2(s+t)^2} \\ &= \frac{2}{s+t} \end{aligned}$$

$$= \frac{2}{s+t}$$

References